

Week 3 - part 1

CS 4200 / 6200

Linear Algebra for QC

Nella Ludlow, PhD
Wright State University
Week 3

Agenda / Map of where we are

- ▶ We covered BIG PICTURE of Quantum Weeks 1 & 2
- ▶ We also covered very top level of Physics of Quantum Mechanics
- ▶ In this week, we will cover Linear Algebra (may be review for some)
 - ▶ This is so we can better understand our Quantum Logic
- ▶ This Video Lecture (W3-part 1) will be Why use Matrices, Matrix Addition, and Scalar Multiplication
- ▶ Next Video Lecture (W3-part 2) and Multiplying Two Matrices, Naïve method, Strassen Method, and other more efficient method for large matrices
- ▶ Final Video Lecture (W3-part 3) will be more about Dirac Notation, How Linear Algebra Describes Quantum Gates, and Tensors

This Week What You Should Learn

- Decide whether two matrices are equal.
- Add and subtract matrices and multiply matrices by scalars.
- Multiply two matrices.
- How to apply a Tensor.
- Use matrix operations to model and solve real-life problems.

Empowering Quantum Jobs: Top 5 Skills for Quantum Coding

- ▶ <https://www.indiatimes.com/technology/news/quantum-coding-top-5-skills-india-jobs-ibm-548748.html>
- ▶ Sep 5, 2021 - India Times
- ▶ The Indian government has called out quantum computing as one of the major technology disruptions that will likely change the entire paradigm of computation, communication, and encryption. It has announced a National Mission on Quantum Technologies & Applications to focus on quantum technologies. While the mission's mandate is broader, a strong component of it is to develop skilled manpower and to develop the industry ecosystem to prepare for this disruption and for India to lead it.
- ▶ (This is good information even if you're not from India - Quantum computing is growing all over the world - China is in the lead)



Budget 2020 announced Rs 8,000 crore over the next 5-yr in the National Mission on Quantum technology and its applications

- The areas of focus for the NM-QTA Mission will be in fundamental science, translation, technology development and towards addressing issues concerning national priorities
- The mission can help prepare next generation skilled manpower, boost translational research and also encourage entrepreneurship and start-up ecosystem development.
- Quantum principles will be used for engineering solutions to extremely complex problems in computing, communications, sensing, chemistry, cryptography, imaging and mechanics

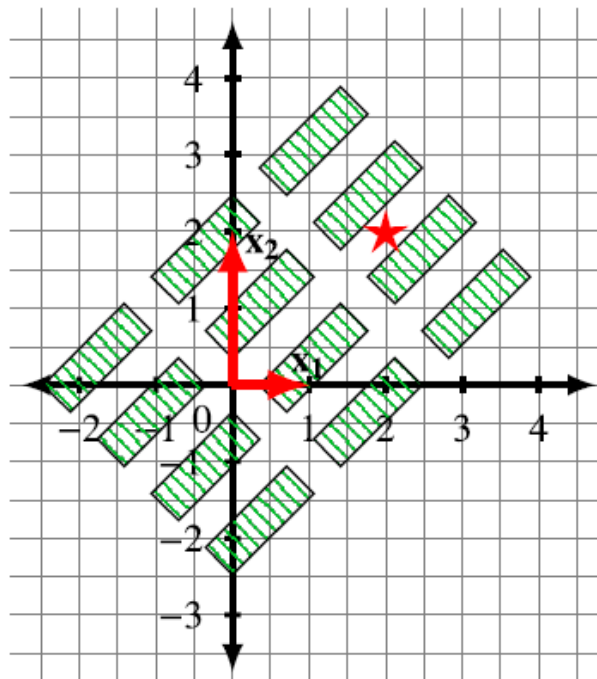
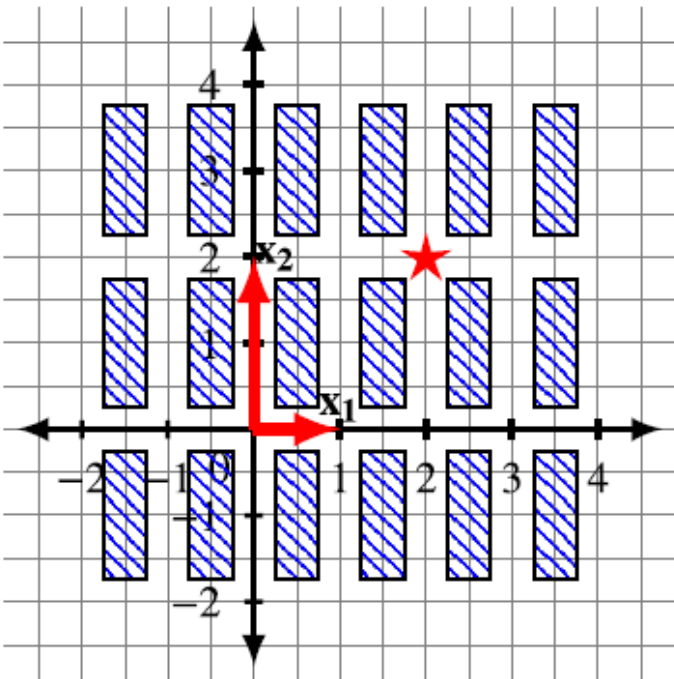


- Their applications which will be boosted include those in aero-space engineering, numerical weather predictions, simulations, securing the communications & financial transactions, cyber security, advanced manufacturing, health, agriculture, education
- It can bring India in the list of few countries with an edge in this emerging field will have a greater advantage in garnering multifold economic growth and dominant leadership role

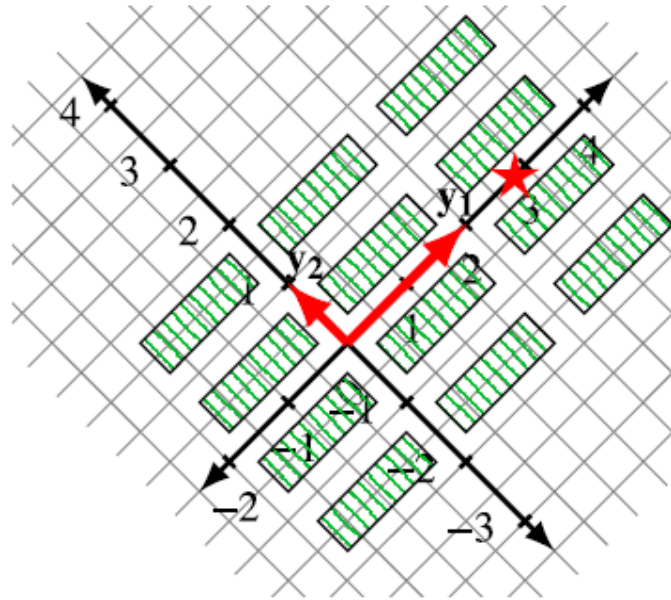
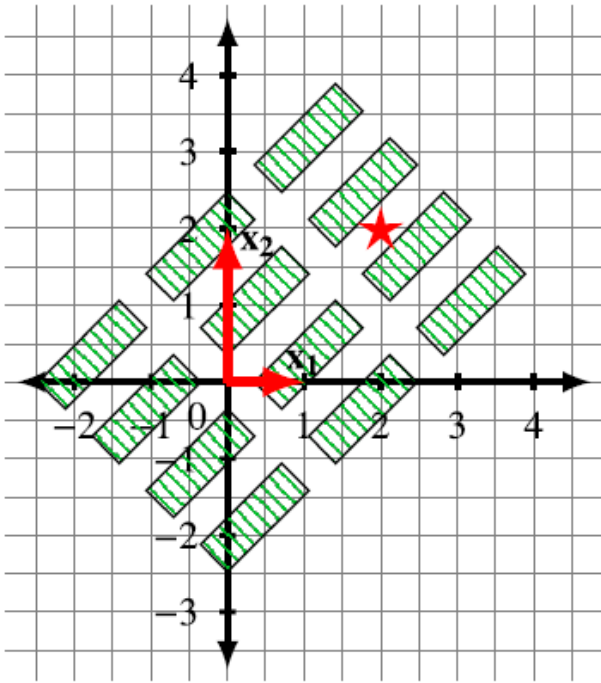
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Professor Nella Ludlow - Wright State University

Why Matrices?



Create a New Coordinate System?

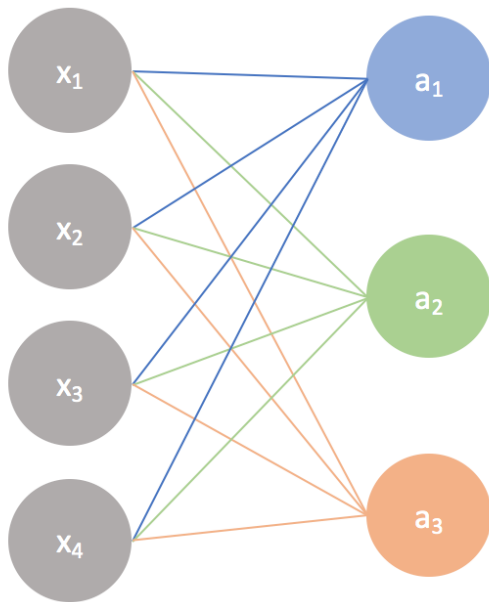


Neural Network is Represented by a Matrix

Input layer

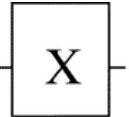
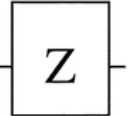
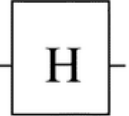
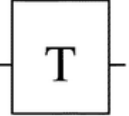
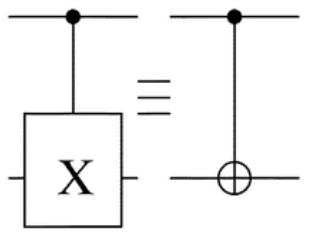
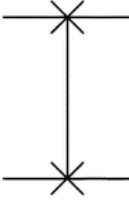
Output layer

A simple neural network



$$\begin{bmatrix} w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} b \\ b \\ b \end{bmatrix} = \begin{bmatrix} w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \end{bmatrix} \xrightarrow{\text{activation}} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

Quantum may be the ultimate Matrix Multiplication / Tensor Machine

X Gate Bit-flip, Not		$\equiv$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$	$=$	$\beta 0\rangle + \alpha 1\rangle$
Z Gate Phase-flip		$\equiv$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$	$=$	$\alpha 0\rangle - \beta 1\rangle$
H Gate Hadamard		$\equiv$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$	$=$	$\frac{\alpha+\beta 0\rangle + \alpha-\beta 1\rangle}{\sqrt{2}}$
T Gate		$\equiv$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$	$=$	$\alpha 0\rangle + e^{i\pi/4}\beta 1\rangle$
Controlled Not Controlled X CNot		$\equiv$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$	$=$	$a 00\rangle + b 01\rangle + d 10\rangle + c 11\rangle$
Swap		$\equiv$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$	$=$	$a 00\rangle + c 01\rangle + b 10\rangle + d 11\rangle$



Let's Start here

Equality of Matrices

Today may be a REVIEW

Equality of Matrices

- This section introduces some fundamentals of matrix theory. It is standard mathematical convention to represent matrices in any of the following three ways.

Representation of Matrices

1. A matrix can be denoted by an uppercase letter such as

A , B , or C .

2. A matrix can be denoted by a representative element enclosed in brackets, such as

$[a_{ij}]$, $[b_{ij}]$, or $[c_{ij}]$.

3. A matrix can be denoted by a rectangular array of numbers such as

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdot & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdot & \cdot & \cdot & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdot & \cdot & \cdot & a_{3n} \\ \vdots & \vdots & \vdots & & & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdot & \cdot & \cdot & a_{mn} \end{bmatrix}.$$

Equality of Matrices

- ▶ Two matrices
- ▶ $A = [a_{ij}]$ and $B = [b_{ij}]$
- ▶ are **equal** when they have the same dimension ($m \times n$) and all of their corresponding entries are equal.

Example 1 - *Equality of Matrices*

- ▶ Solve for a_{11} , a_{12} , a_{21} , and a_{22} in the following matrix equation.

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 0 \end{bmatrix}$$

- ▶ **Solution:**
- ▶ Because two matrices are equal only when their corresponding entries are equal, you can conclude that
- ▶ $a_{11} = 2$, $a_{12} = -1$, $a_{21} = -3$, and $a_{22} = 0$.

Equality of Matrices

- ▶ Be sure you see that for two matrices to be equal, they must have the same dimension *and* their corresponding entries must be equal.
- ▶ For instance,

$$\begin{bmatrix} 2 & -1 \\ \sqrt{4} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 2 & 0.5 \end{bmatrix} \quad \text{but} \quad \begin{bmatrix} 2 & -1 \\ 3 & 4 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}.$$

Matrix Addition and Scalar Multiplication

Matrix Addition and Scalar Multiplication

- ▶ You can add two matrices (of the same dimension) by adding their corresponding entries.

Definition of Matrix Addition

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrices of dimension $m \times n$, then their sum is the $m \times n$ matrix given by

$$A + B = [a_{ij} + b_{ij}].$$

The sum of two matrices of different dimensions is undefined.

Example 2 - *Addition of Matrices*

- ▶ a.
$$\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -1 + 1 & 2 + 3 \\ 0 + (-1) & 1 + 2 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix}$$
- ▶ b.
$$\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
- ▶ c. The sum of $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix}$
- ▶ is undefined because A is of dimension 2×3 and B is of dimension 2×2 .

Matrix Addition and Scalar Multiplication

- In operations with matrices, numbers are usually referred to as **scalars**. In this text, scalars will always be real numbers. You can multiply a matrix A by a scalar c by multiplying each entry in A by c .

Definition of Scalar Multiplication

If $A = [a_{ij}]$ is an $m \times n$ matrix and c is a scalar, then the **scalar multiple** of A by c is the $m \times n$ matrix given by

$$cA = [ca_{ij}].$$

Example 3 - *Scalar Multiplication*

- For the following matrix, find $3A$.

$$A = \begin{bmatrix} 2 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix}$$

- **Solution:**

$$3A = 3 \begin{bmatrix} 2 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 3(2) & 3(2) & 3(4) \\ 3(-3) & 3(0) & 3(-1) \\ 3(2) & 3(1) & 3(2) \end{bmatrix} = \begin{bmatrix} 6 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix}$$

Matrix Addition and Scalar Multiplication

- ▶ The properties of matrix addition and scalar multiplication are similar to those of addition and multiplication of real numbers.
- ▶ One important property of addition of real numbers is that the number 0 is the additive identity.
- ▶ That is, $c + 0 = c$ for any real number. For matrices, a similar property holds.

Matrix Addition and Scalar Multiplication

- ▶ For matrices, a similar property holds. That is, if A is an $m \times n$ matrix and O is the $m \times n$ **zero matrix** consisting entirely of zeros, then $A + O = A$.
- ▶ In other words, O is the **additive identity** for the set of all $m \times n$ matrices. For example, the following matrices are the additive identities for the sets of all 2×3 and 2×2 matrices.

- ▶ $O = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{2 \times 3 \text{ zero matrix}} \quad \text{and} \quad O = \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}_{2 \times 2 \text{ zero matrix}}$

Matrix Addition and Scalar Multiplication

Properties of Matrix Addition and Scalar Multiplication

Let A , B , and C be $m \times n$ matrices and let c and d be scalars.

1. $A + B = B + A$

Commutative Property of Matrix Addition

2. $A + (B + C) = (A + B) + C$

Associative Property of Matrix Addition

3. $(cd)A = c(dA)$

Associative Property of Scalar Multiplication

4. $1A = A$

Scalar Identity

5. $A + O = A$

Additive Identity

6. $c(A + B) = cA + cB$

Distributive Property

7. $(c + d)A = cA + dA$

Distributive Property

Example 5 - *Using the Distributive Property*

$$3\left(\begin{bmatrix} -2 & 0 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} 4 & -2 \\ 3 & 7 \end{bmatrix}\right) = 3\begin{bmatrix} -2 & 0 \\ 4 & 1 \end{bmatrix} + 3\begin{bmatrix} 4 & -2 \\ 3 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & 0 \\ 12 & 3 \end{bmatrix} + \begin{bmatrix} 12 & -6 \\ 9 & 21 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -6 \\ 21 & 24 \end{bmatrix}$$

Matrix Addition and Scalar Multiplication

- ▶ The algebra of real numbers and the algebra of matrices have many similarities. For example, compare the following solutions.

- ▶ *Real Numbers*

- ▶ *$m \times n$ Matrices*

- ▶ *(Solve for x .)*

- ▶ *(Solve for X .)*

- ▶ $x + a = b$

- ▶ $X + A = B$

- ▶ $x + a + (-a) = b + (-a)$

- ▶ $X + A + (-A) = B + (-A)$

- ▶ $x + 0 = b - a$

- ▶ $X + O = B - A$

- ▶ $x = b - a$

- ▶ $X = B - A$

Example 6 - *Solving a Matrix Equation*

- ▶ Solve for X in the equation

$$3X + A = B$$

- ▶ where

- ▶ $A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix}$.

Example 6 - *Solution*

► Begin by solving the equation for X to obtain

► $3X = B - A$

► $X = \frac{1}{3}(B - A)$

► Now, using the A matrices and B , you have

►
$$X = \frac{1}{3} \left(\begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} \right)$$

$$= \frac{1}{3} \begin{bmatrix} -4 & 6 \\ 2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{4}{3} & 2 \\ \frac{2}{3} & -\frac{2}{3} \end{bmatrix}$$

Substitute the matrices

Subtract matrix A from matrix B

Multiply the resulting matrix by $\frac{1}{3}$

Algorithm Complexity - Matrix Addition

Where are we on this scale?

1-1 Comparison of running times

For each function $f(n)$ and time t in the following table, determine the largest size n of a problem that can be solved in time t , assuming that the algorithm to solve the problem takes $f(n)$ microseconds.

	second	minute	hour	day	month	year	century
$\lg n$							
$\sqrt{n}$							
n							
$n \lg n$							
n^2							
n^3							
2^n							
$n!$							



Fastest!

Fast!

Linear!

Merge Sort->

Most Sorts->

Slow!

Super slow!

Intractable!

Algorithm Complexity - Matrix Scalar Multiplication

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What's in Next Video

- ▶ Matrix Multiplication